

Smart Plant Carer · Premium Course

2 live sessions · 2 hrs each · 9-13 yrs · from KES 7,000

Premium course outline **PREMIUM**

Overview

Build a self-watering planter that senses dry soil and pumps water to a real seedling all by itself.

DETAIL

Track	Arduino Robotics & Electronics
Format	2 live sessions · 2 hrs each
Delivery	Live online (1:1 or small group) or in person, you choose when booking
Ages	9-13 yrs
Level	Beginner → Intermediate
Price	from KES 7,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16x2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Needed at home for live online sessions; for in-person sessions we provide kits, so buy one only to keep building at home.
Parts pack	Optional, KES 1,000: Soil-moisture sensor, Mini water pump + tubing + reservoir, Pot and seedling
Standalone	Yes
Prerequisites	None — standalone

Programme

1. **Sense & pump.** Wire a soil-moisture sensor and calibrate the dry vs. wet readings on the Serial Monitor, then drive a small water pump through a relay module.

2. **Automatic planter.** Set a moisture threshold so it waters only when the soil is dry, then assemble the pump, tubing and reservoir around a potted seedling and run a full automatic watering cycle.

What they take home

 Self-watering planter kit + a seedling

Why parents love it

"They build tech that keeps a plant alive on its own — and understand every part of why it works."

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Smart Plant Carer**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.